

**Summary of NFPA Inspection, Test, and Maintenance Requirements
Clean Agent Systems (FM-200, HFC-227)**

Inspections, tests, and maintenance of clean agent fire extinguishing systems shall be performed in accordance with Unified Facilities Criteria (UFC) 3-601-02, the system manufacturer's instructions, and NFPA 2001 (latest edition). The following list highlights the minimum requirements, however, this list is not meant to replace manufacturer's instructions and updated code requirements.

Monthly and annual maintenance tags shall be attached to each fire suppression system for recording the inspector's initials, date, and confirmation on maintenance/inspections performed. Where fusible links are used, the manufacture and the installation dates for the links shall be marked on the system inspection tag. At a minimum the inspector(s) shall be licensed as a NICET Level II and the supervisor shall be licensed as a NICET Level III with experience in inspecting clean agent system. The helper(s) are/is not required to be licensed. Five (5) year inspections shall be conducted the first year of contract award.

ITEM	ACTIVITY	PERIODIC FREQUENCY					
		Weekly	Monthly	Quarterly	Semi-Annual	Annual	Every 5 years
System condition and operation	Inspection/Test					✓	
Check agent quantity and pressure.	Inspection				✓		
Halocarbon Clean Agent Cylinders that show more than 5% net loss in agent weight or loss in pressure (adjusted for temperature) of more than 10% shall be refilled or replaced	Inspection				✓		
Inert gas clean agent cylinders containing agent under pressure (not liquified) showing a loss in pressure (adjusted for temperature) of more than 5% shall be refilled or replaced	Inspection				✓		
Cylinders shall not be recharged without retesting if more than 5 years have elapsed since the date of the last test and inspection. Hydrostatic pressure test agent containers, auxiliary pressure containers, and hose assemblies.	Test						✓
Cylinders in continuous operation without discharging	Inspection						✓
Examine cylinders for damage or corrosion	Inspection				✓		
Examine system hoses for damage	Inspection					✓	
System hoses	Test						✓
Room enclosure – ensure penetrations are protected	Inspection				✓		
Piping and nozzles	Inspection				✓		
Check hydrostatic test date for cylinders and discharge hoses	Inspection					✓	
Agent storage containers properly located	Inspection				✓		
Containers and mounting brackets	Inspection				✓		
Audible and visual alarms	Test					✓	
System alarm, detection and release controls	Test					✓	
Clean detectors	Maintenance					✓	

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ITEM	ACTIVITY	PERIODIC FREQUENCY					
		Weekly	Monthly	Quarterly	Semi-Annual	Annual	Every 5 years
Remote monitoring operation - verify receipt of condition at the remote panel	Test				✓		
Check system control unit and battery backup in accordance with NFPA 72	Test					✓	
Discharge - test the system when maintenance/ Authority Having Jurisdiction (AHJ) indicates their advisability	Test					✓	

**Summary of NFPA Inspection, Test, and Maintenance
Carbon Dioxide (CO2) Systems**

Inspections, tests, and maintenance of carbon dioxide extinguishing systems shall be performed in accordance with Unified Facilities Criteria (UFC) 3-601-02, the system manufacturer's instructions, and NFPA 12 (latest edition). The following list highlights the minimum requirements, however, this list is not meant to replace manufacturer's instructions and updated code requirements.

Monthly and annual maintenance tags shall be attached to each fire suppression system for recording the inspector's initials, date, and confirmation on maintenance/inspections performed. Where fusible links are used, the manufacture and the installation dates for the links shall be marked on the system inspection tag. At a minimum the inspector(s) shall be licensed as a NICET Level II and the supervisor shall be licensed as a NICET Level III with experience in inspecting clean agent system. The helper(s) are/is not required to be licensed. Five (5) year inspections shall be conducted the first year of contract award.

ITEM	ACTIVITY	PERIODIC FREQUENCY						
		Weekly	Monthly	Quarterly	Semi-Annual	Annual	Every 5 years	Every 12 years
Low-Pressure Containers liquid level gauge	Inspection				✓			
System operational condition	Inspection				✓			
High Pressure Cylinders	Weigh				✓			
System hoses	Test						✓	
System operation	Test					✓		
Low pressure CO2 tank, check for signs of damage or corrosion, verify amount of CO2 in tank	Inspection					✓		
High pressure cylinders in place, secure, free of damage or corrosion. Verify hydrostatic test date	Inspection					✓		
Hazard size, type, and configuration is unchanged	Inspection					✓		
System time delays	Test					✓		
Audible and visible alarms	Test					✓		
Warning signs properly installed and displayed	Inspection					✓		
System detectors	Test					✓		
Room enclosure – ensure penetrations are protected	Inspection				✓			
High-Pressure Cylinders that show more than 10% net loss in agent weight shall be refilled or replaced	Inspection				✓			
Low-Pressure Container that shows more than 10% net loss in agent shall be refilled unless amount of agent meets minimum gas requirements	Inspection				✓			
All system isolation valves are in their normal open positions	Inspection/Test					✓		
Clean initiation devices (heat, smoke, duct detectors, etc.)	Maintenance					✓		
System pressure release devices (dampers, door closures, etc.) operate fully	Test					✓		
Selector valves operate and reset properly	Test					✓		
Equipment shutdowns operate correctly	Test					✓		
Manual release operates correctly and within specifications	Test					✓		
System piping is secure, free of damage, properly supported and connected to the CO2 supply	Inspection					✓		
Nozzles are free of damage, properly supported, oriented and free of contamination	Inspection					✓		
Check system control unit and battery backup in accordance with NFPA 72	Test					✓		

**Summary of NFPA Inspection, Test, and Maintenance
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ITEM	ACTIVITY	PERIODIC FREQUENCY						
		Weekly	Monthly	Quarterly	Semi-Annual	Annual	Every 5 years	Every 12 years
Emergency manual control present, accessible, labeled and operable	Inspection					✓		
High pressure cylinders shall not be recharged without a hydrostatic test and remarking if more than 5 years have elapsed from the date of the last test	Test						✓	
High pressure cylinders and hose assemblies shall be discharged, hydrostatically tested and reinstalled	Test						✓	
Discharge - test the system when maintenance/ Authority Having Jurisdiction (AHJ) indicates their advisability	Test					✓		

Summary of NFPA Inspection, Test, and Maintenance Requirements for Kitchen Ventilation Hoods

Inspections, tests, and maintenance on kitchen ventilation hoods and fire suppression systems shall be performed in accordance with Unified Facilities Criteria (UFC) 3-601-02, the manufacturer's instructions, NFPA 17 (latest edition), NFPA 17A (latest edition), NFPA 72 (latest edition), and NFPA 96 (latest edition). The following list highlights the minimum requirements, however, this list is not meant to replace manufacturer's instructions and updated code requirements.

Monthly and annual maintenance tags shall be attached to each fire suppression system for recording the inspector's initials, date, and confirmation on maintenance/inspections performed. Where fusible links are used, the manufacture and the installation dates for the links shall be marked on the system inspection tag. At a minimum the inspector(s) shall be licensed as a NICET Level II and the supervisor shall be licensed as a NICET Level III with experience in inspecting clean agent system. The helper(s) are/is not required to be licensed. Five (5) year inspections shall be conducted the first year of contract award.

ITEM	ACTIVITY	PERIODIC FREQUENCY					
		Weekly	Monthly	Quarterly	Semi-Annual	Annual	Every 5 years
Extinguishing system nozzles are located directly above grease producing equipment	Inspection				✓		
The Manual actuators are unobstructed	Inspection				✓		
The tamper indicators and seals are intact	Inspection				✓		
The maintenance tag or certificate is in place	Inspection				✓		
No obvious physical damage or condition exists that might prevent operation	Inspection				✓		
The pressure gauge(s) are in operable range	Inspection				✓		
The nozzle blow-off caps are intact and undamaged	Inspection				✓		
Neither protected equipment nor hazard has been replaced, modified, or relocated	Inspection				✓		
Gas and electric power shutoff are operational	Test				✓		
Water-wash hood cleaning systems are operational in conjunction with hoods protected by sprinkler systems	Test				✓		
Remove grease from exhaust system	Maintenance				✓		
Fixed temperature sensing elements of the fusible alloy type	Replace				✓		
Clean fixed temperature sensing elements other than the fusible alloy type	Maintenance					✓	
Restorable-type Heat detectors condition for damage or buildup of foreign debris	Maintenance					✓	

Summary of NFPA Inspection, Test, and Maintenance Requirements for Kitchen Ventilation Hoods

ITEM	ACTIVITY	PERIODIC FREQUENCY					
		Weekly	Monthly	Quarterly	Semi-Annual	Annual	Every 5 years
Restorable-type Heat detectors operational/functional test	Maintenance					✓	
Restorable-type Heat detector calibration verification test	Maintenance					✓	
Nonrestorable Heat detectors functional test	Maintenance					✓	
Manual release stations are operational and send a signal to the building fire alarm control panel	Test				✓		
Automatic release devices are operational and send a signal to the building fire alarm control panel	Test				✓		
Water flow, valve tamper, and low water pressure cutoffs are operational	Test				✓		
Verify the agent distribution piping is not obstructed	Test				✓		
Examine detectors, expellant gas containers, agent containers, releasing devices, piping, hose assemblies, nozzles, signals, and all auxiliary equipment	Maintenance				✓		
Hydrostatic pressure test agent containers, auxiliary pressure containers, and hose assemblies	Test						✓
Recirculating systems operation and safety interlocks perform in accordance with manufacturer's instructions	Test				✓		
Recirculating systems - Clean electrostatic precipitators	Maintenance				✓		
Recirculating systems - Clean entire hood Plenum and blower section	Maintenance				✓		